Yuxi Chen

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School of Mathematical Sciences, University of Science and Technology of China (USTC)

Education

Sep. 2026 – Present — **Ph.D. Student in Mathematics**, School of Mathematical Sciences, University of Science and Technology of China (USTC). Expected completion: 2031.
Supervisor: Prof. Juyong Zhang.

Sep. 2022 – Jun. 2026 — **B.Sc. in Mathematics**, major in Information and Computing Science, University of Science and Technology of China (USTC).

Research Experience

Sep. 2026 – Present — **Exploring Expressive 3D Facial Representations**
University of Science and Technology of China

- Exploring 3D facial representations capable of capturing fine-grained facial geometry and diverse expression variations.
- Investigating geometry-based optimization and deformation modeling, with experiments to evaluate representation capacity and reconstruction quality.

Selected Projects

Oct. 2025 – May. 2026 — **Single-Image Audio-Driven Avatar Generation with SyncNet-Based Lip-Synchronization Refinement**
Undergraduate Thesis | University of Science and Technology of China
Supervisor: Prof. Juyong Zhang

- Worked with an existing two-stage audio-driven avatar generation framework, consisting of a pretrained Audio2FLAME model for predicting FLAME expression coefficients from driving audio and a FLAME2Pixel network for reference-image-conditioned video synthesis.
- Integrated SyncNet-based audio-visual synchronization supervision to fine-tune the FLAME2Pixel network, exploring an auxiliary objective for improving lip synchronization.

May. 2025 — **Incremental Sparse Structure-from-Motion (SfM) Reconstruction**
Mathematical Modeling Course Project | University of Science and Technology of China

- Implemented camera calibration, SIFT feature extraction, FLANN matching, and RANSAC-based relative pose estimation for sparse 3D reconstruction.
- Developed triangulation, incremental camera registration with PnP-RANSAC, and Ceres-based bundle adjustment; visualized reconstructed point clouds using Open3D.

Research Interests

- 3D Facial Geometry Reconstruction and Deformation Modeling
- 3D Human Representation and Controllable Avatar Generation
- Generative Models for Human and Video Generation
- Multimodal AI and Interactive Avatars

Selected Coursework

GRADUATE COURSEWORK

- Digital Image Processing — 91/100
- Geometric Deep Learning — 85/100
- Numerical Methods for PDEs — 92/100

UNDERGRADUATE COURSEWORK

- Computer Graphics — 94/100
- Frontiers of Computer Graphics — 94/100
- Mathematical Modeling — A(4.0/4.3)
- Probability Theory — 92/100
- Mathematical Statistics — 95/100

Awards & Honors

2025 — Silver Award, USTC Outstanding Student Scholarship
2022 — USTC Outstanding Freshman Scholarship